Time-series Prediction of Heat Flux Traces in Tokamak Reactors

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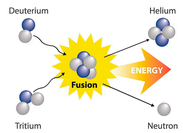
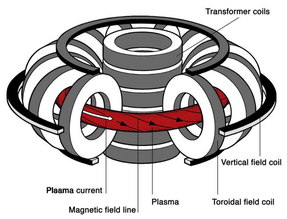
Finetuned time-series foundation models **outperform neural surrogate models** on forecasting gyrokinetic flux.

Introduction



Nuclear Fusion [8]



Tokamak Schematic [9]

Energy (Heat) Flux

- physically and practically relevant observable
- determines how much energy is transported out of the plasma - and therefore overall plasma confinement and reactor performance

Neural Surrogate Models [1]

- replace expensive nonlinear gyrokinetic simulations by modeling 5D plasma turbulence (Table 3)
- trade detailed turbulence information for fast and scalable heat flux predictions

Motivation

For practical use, **efficiency is the dominant requirement**. [11]

Probabilistic Forecasting Foundation Models [4-7]

- can predict time-resolved heat flux trajectories
- neglect physical structure and interpretability
- can be conditioned on physical operating parameters

Research Questions

RQ1: While sacrificing interpretability, can probabilistic time-series foundation models **match or outperform neural surrogate models** for gyrokinetic heat flux prediction?

RQ2: How does **operating-parameter conditioning** affect the performance and robustness of finetuned time-series foundation models for heat flux forecasting?

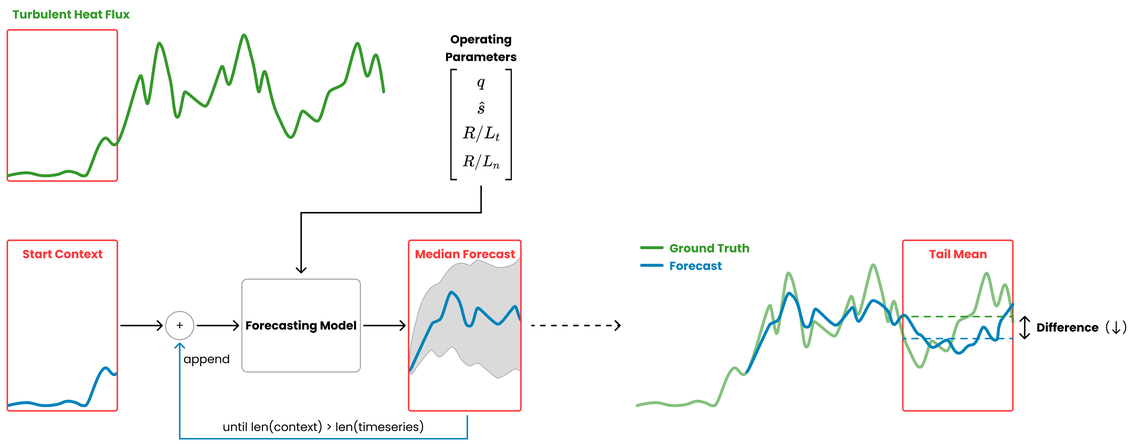
Experiments

- Full and LoRA finetuning [3] with and without operating parameter conditioning (Table 1)
- Zero-shot performance (Table 2)

Evaluation: RMSE of the **average heat flux over the last 80 timesteps** of a simulated flux trace using an **autoregressive rollout** starting from the first 80 timesteps.

Dictionary

q	→	Safety Factor
$\hat{s}$	→	Magnetic Sheering
R/L_t	→	Ion Temperature Gradient
R/L_n	→	Density Gradient
$\odot$	→	Hadamard Product: pointwise multiplication
$\bar{Q}$	→	Average Heat Flux
$\sigma(\|\|\cdot\|\|^2)$	→	Sigmoid Gate on squared vector norm
$Emb(\cdot)$	→	Continuous Sin-Cos Frequency Embedding [10]



Base Model	Finetuning Type	$\bar{Q}$ ID (↓)	$\bar{Q}$ OOD	Trainable Params %	Trainable Params #Mio.	Inference Time [s]
google/timesfm-2.0-500m	Full Finetuning*	20.67 ± 7.43	12.01 ± 3.21	100.0	498.8	0.091 ± 1.65e-3
google/timesfm-2.0-500m	BilinearLoRA	20.15 ± 7.79	7.11 ± 1.32	1.22	6.2	0.245 ± 2.17e-3
google/timesfm-2.0-500m	OSSBilinearLoRA	19.24 ± 7.87	7.74 ± 2.08	28.91	202.8	0.291 ± 3.85e-3
GyroSwin-1B [1]	-	18.35 ± 1.56	26.43 ± 9.49	100.0	1000.0	2.849**
google/timesfm-2.0-500m	RSSBilinearLoRA	18.03 ± 6.81	7.86 ± 2.20	1.39	7.0	0.304 ± 2.50e-3
google/timesfm-2.0-500m	LoRA*	17.76 ± 8.05	16.07 ± 4.18	1.02	5.1	0.081 ± 1.51e-3
amazon/chronos2	LoRA*	16.73 ± 6.67	5.08 ± 1.22	1.0	1.2	0.067 ± 2.95e-2
amazon/chronos2	RSSBilinearLoRA	16.33 ± 5.39	5.65 ± 2.03	1.86	2.3	0.170 ± 6.26e-3
amazon/chronos2	OSSBilinearLoRA	16.11 ± 6.18	**3.19 ± 0.73**	25.0	39.8	0.159 ± 4.59e-3
amazon/chronos2	Full Finetuning*	15.50 ± 4.47	4.76 ± 0.89	100.0	119.5	0.050 ± 7.07e-4
amazon/chronos2	BilinearLoRA	**13.83 ± 4.18**	4.86 ± 0.68	1.54	1.9	0.136 ± 8.64e-4

Table 1 : Comparison of the finetuned performance across base models and GyroSwin. For average heat flux $\bar{Q}$ we report RMSE of time-averaged predictions after an autoregressive rollout. Time-series models are trained and benchmarked on a NVIDIA RTX 4070 16GB Ti Super.
(*) No operating parameter conditioning.
(**) To compare the inference speed to GyroSwin we use the reported 15.4ms forward pass inference speed and multiply by the number of rollout steps (185). The large speed gap can be mainly attributed to the fact that time-series models do forecast 64 timesteps in one forward-pass. GyroSwin was benchmarked on a NVIDIA H100 80GB HBM3.

Base Model	$\bar{Q}$ ID (↓)	$\bar{Q}$ OOD	Inference Time [s]
google/timesfm-2.0-500m	156.17 ± 67.31	98.61 ± 23.55	5.65 ± 5.82e-2
amazon/chronos-bolt-tiny	110.15 ± 14.08	92.89 ± 21.16	**0.030 ± 1.08e-3**
amazon/chronos2	107.09 ± 15.74	87.47 ± 22.08	0.073 ± 1.72e-3
google/timesfm-2.5-200m	104.23 ± 14.87	87.20 ± 25.05	0.231 ± 7.23e-3
NX-AI/TiRex	**79.49 ± 14.38**	**64.03 ± 19.53**	1.95 ± 1.63e-2

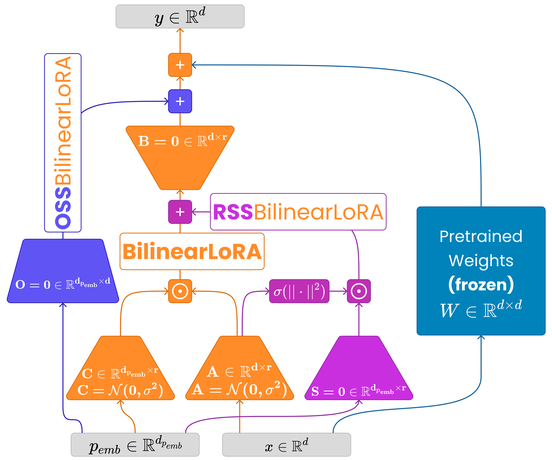
Table 2 : Zero-shot performance across five time-series foundation models.

Method	Average Flux	Diagnostics	Zonal Flows	Turbulence
Time-series models	1D→0D	X	X	X
Tabular Regressors, e.g., GPR, MLP	1D→0D	X	X	X
SOTA Reduced Numerical modelling, e.g., QL	3D→0D	3D→1D	X	X
Neural Surrogates, e.g., GyroSwin [1]	5D→0D	5D→1D	5D→1D	5D→5D

Table 3 : Comparison of different modeling approaches by capabilities [1] extended by time-series models.

Methodology

Inspired by Feature-wise Linear Modulation (FiLM) [2] we modify low-rank adapters to condition a model on physical operating parameters.



Standard LoRA

$$y = Wx + \frac{\alpha}{r} B(Ax)$$

BilinearLoRA (Bi-Linear Low-Rank Adaption)

Feature-wise bilinear interaction between input features and operating parameter embeddings in rank space.

$$y = Wx + \frac{\alpha}{r} B\Big((Ax) \odot (Cp_{emb})\Big), p_{emb} = Emb(p)$$

RSSBilinearLoRA (**R**ank-**S**pace-**S**hift BilinearLoRA)

FiLM via multiplicative and additive transformation in rank space.

$$p_{emb} = Emb(p), h = Ax, g_s(h) = \sigma(||h||^2)$$

$$y = Wx + \frac{\alpha}{r} B\Big(h \odot (Cp_{emb}) + (Sp_{emb}) \odot g_s(h)\Big)$$

OSSBilinearLoRA (**O**utput-**S**pace-**S**hift BilinearLoRA)

Feature-wise bilinear modulation via multiplicative transformation in rank space and additive transformation in output space.

$$y = Wx + \frac{\alpha}{r}\Big(B\big((Ax) \odot (Cp_{emb})\big)\Big) + \frac{1}{\sqrt{0.5 \cdot d_{p_{emb}}}}(Op_{emb})$$

Results and Conclusion

Finetuned (Table 1) and Zero-shot (Table 2) performance of probabilistic time-series foundation models on turbulent heat flux traces.

→ **Zero-shot forecasting** matches neural surrogate model performance with **TiRex** showing superior results.

→ **Finetuning** outperforms best neural surrogate model on ID and OOD benchmarks.

→ Probabilistic Time-Series Foundation Models do benefit from physical operating parameter conditioning but need more optimization.

→ Complete rollout inference is up to **60 times faster** than GyroSwin.

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